Introduction to the System Function and System Poles and Zeros

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Key Concepts

1) The system function $H(z)$ is a generalization of the frequency response description of a system, $H(e^{j\omega})$.

    a) Frequency response $H(e^{j\omega})$ describes how the system changes a complex sinusoid input of frequency ω. If the input $x[n] = e^{j\omega n}$, then the output $y[n] = H(e^{j\omega})x[n]$.

    b) System function $H(z)$ describes how the system changes a complex exponential input $x[n] = z^n = r^n e^{j\omega n}$ where $z = re^{j\omega}$. Thus, if the input $x[n] = z^n$, then the output $y[n] = H(z)x[n]$.

c) The system function is equal to the frequency response for $r = 1$: $H(e^{j\omega}) = H(z)\Big|_{z=e^{j\omega}}$.

2) The value $z = e^{j\omega}$ is located at radius 1 and angle ω in the complex z-plane, so frequency response is obtained by varying ω from 0 to 2π radians, that is, by evaluating the system function on the unit circle $|z| = 1$.

3) The system function for a linear constant coefficient difference equation is a rational function.

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \longleftrightarrow H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

4) Poles are the roots of the denominator of $H(z)$ so at a pole $H(z) \rightarrow \infty$.

5) Zeros are the roots of the numerator of $H(z)$ so at a zero $H(z) = 0$.

6) The poles and zeros of a system tell us a lot about the system behavior

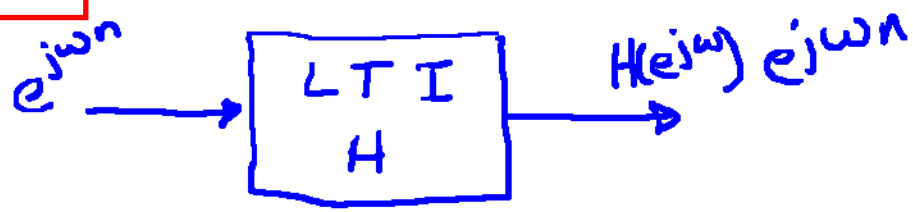
7) The frequency response at ω is proportional to the product of the distances from $e^{j\omega}$ to the zeros divided by the product of the distances from $e^{j\omega}$ to the poles.

a) Zeros located near/on the unit circle pull the frequency response down near/at the zero.

b) Poles located near the unit circle push the frequency response up near the pole.

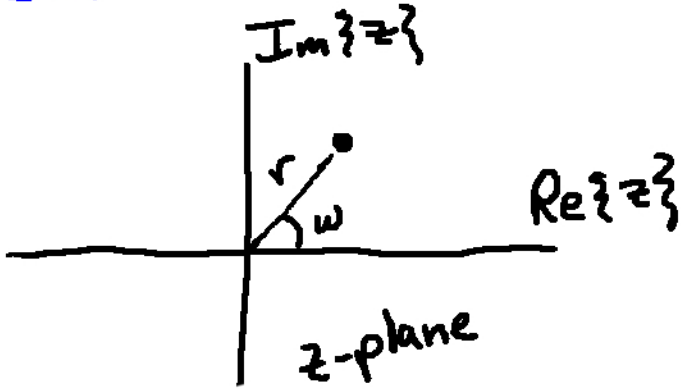
Video Screenshots

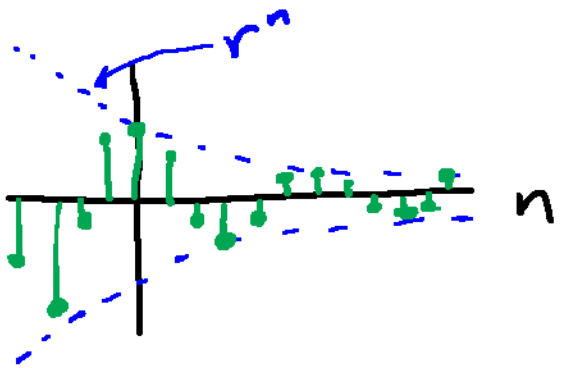
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$H(e^{j\omega})$: frequency response

Let $z = re^{j\omega}$ so $z^n = r^n e^{j\omega n}$



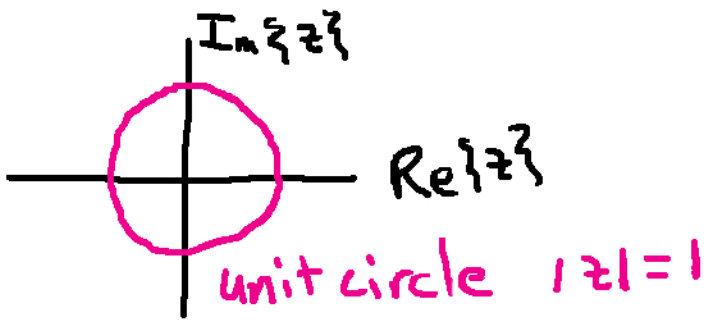




$H(z)$: system function

If $r=1$, $z^n = e^{j\omega n}$ $(z = e^{j\omega})$

$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}}$



$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= z^n \underbrace{\sum_{k=-\infty}^{\infty} h[k]\, z^{-k}}_{\text{z-transform of } h[k] \quad H(z)}$$

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$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \Rightarrow H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

"rational"

Poles: $H(z) \to \infty$ $\Leftarrow$ $\sum_{k=0}^{N} a_k z^{-k} = 0$, $z=0$ if $b_0 = 0$

zeros: $H(z) \to 0$ $\sum_{k=0}^{M} b_k z^{-k} = 0$

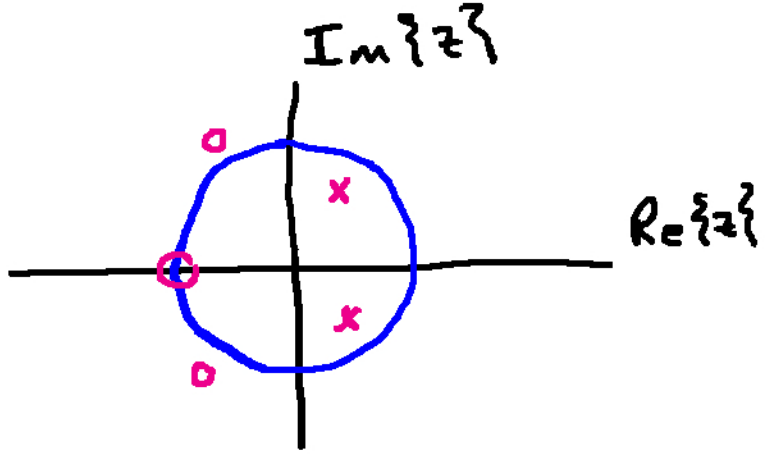
Write $H(z)$ in factored form

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

c_k: zeros, "o"

d_k: poles, "x"

$b_0 \neq 0, a_0 \neq 0$



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Example: $y[n] - \frac{3}{8}y[n-1] - \frac{7}{16}y[n-2] = x[n] + x[n-2]$

Find $H(z)$, poles, zeros.

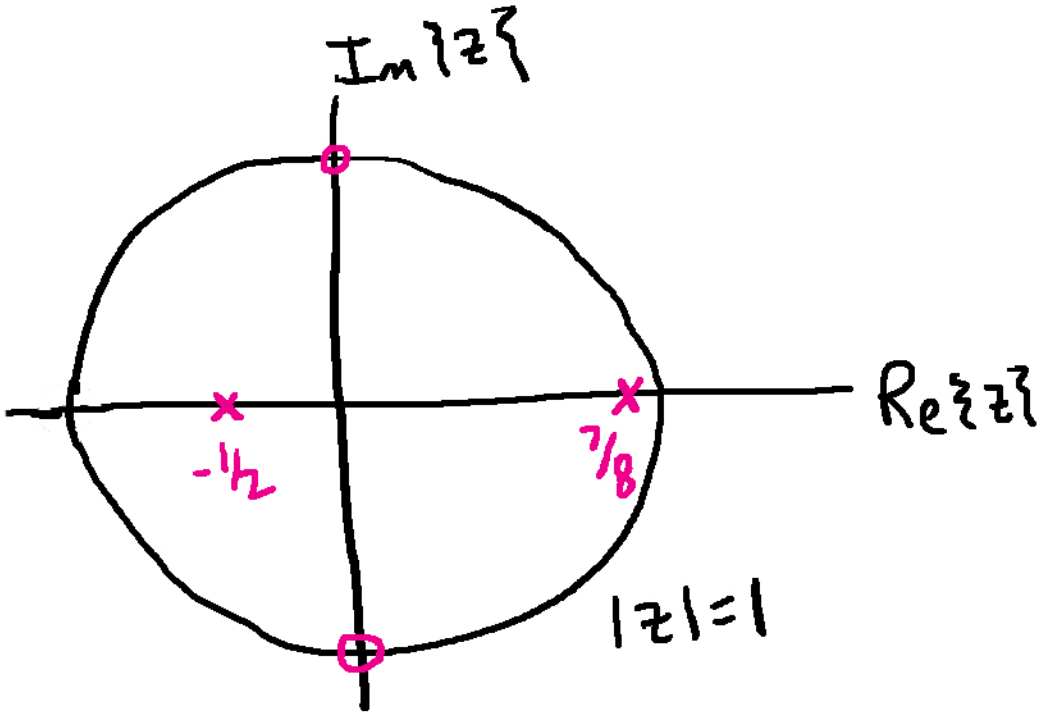
$a_0 = 1,\ a_1 = -3/8,\ a_2 = -7/16 \Rightarrow b_0 = 1,\ b_1 = 0,\ b_2 = 1$

$$H(z) = \frac{1 + z^{-2}}{1 - \frac{3}{8}z^{-1} - \frac{7}{16}z^{-2}} = \frac{(1 + jz^{-1})(1 - jz^{-1})}{(1 - \frac{7}{8}z^{-1})(1 + \frac{1}{2}z^{-1})}$$

zeros: $z = \pm j$

poles: $z = \frac{7}{8},\ z = -\frac{1}{2}$

a_k, b_k real $\Rightarrow$ poles/zeros in conjugate pairs



Pole/zero locations give great insight

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1) Stability - stable (and causal) system $\Leftrightarrow$ all poles inside $|z|=1$ $(|d_k| < 1)$

2) Response time - response of a system at rest at time $n=0$ contains terms: $\sum_{k=1}^{N} \alpha_k d_k^n$ + steady state

transient terms d_k^n

poles near $z=0$ $\Rightarrow$ fast response

poles near $|z|=1$ $\Rightarrow$ slow response

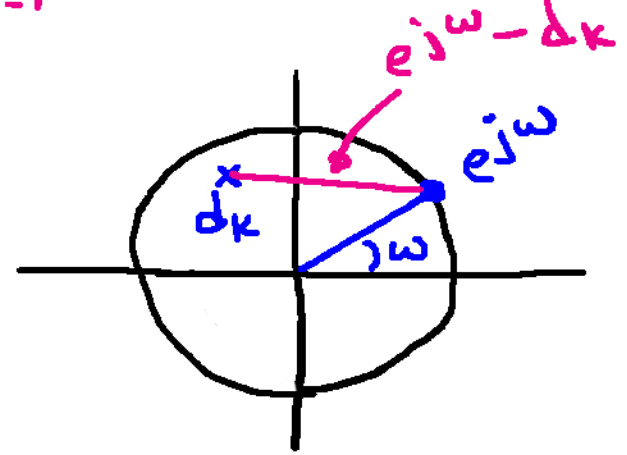
3) Frequency Response - depends on distance between pole/zero locations and unit circle

$H(e^{j\omega}) = H(z)\big|_{z=e^{j\omega}}$ $|z|=1$, the unit circle

$$|H(e^{j\omega})| = \left| \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}(1 - c_k e^{-j\omega})}{\prod_{k=1}^{N}(1 - d_k e^{-j\omega})} \right| = \left|\frac{b_0}{a_0}\right| \frac{\prod_{k=1}^{M}|e^{j\omega} - c_k|}{\prod_{k=1}^{N}|e^{j\omega} - d_k|}$$

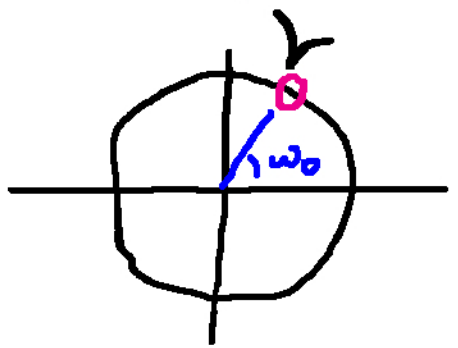
$|e^{j\omega} - c_k|$: distance from $e^{j\omega}$ to c_k

$|e^{j\omega} - d_k|$: distance from $e^{j\omega}$ to d_k

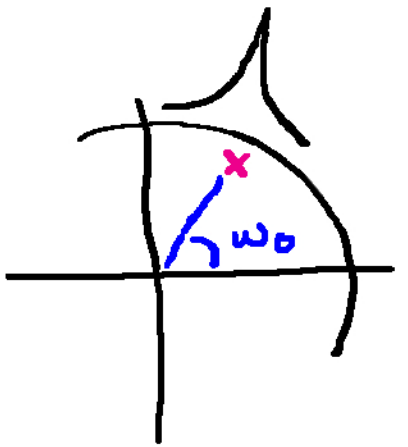


$$|H(e^{j\omega})| = \frac{\prod \text{distances to zeros}}{\prod \text{distances to poles}}$$

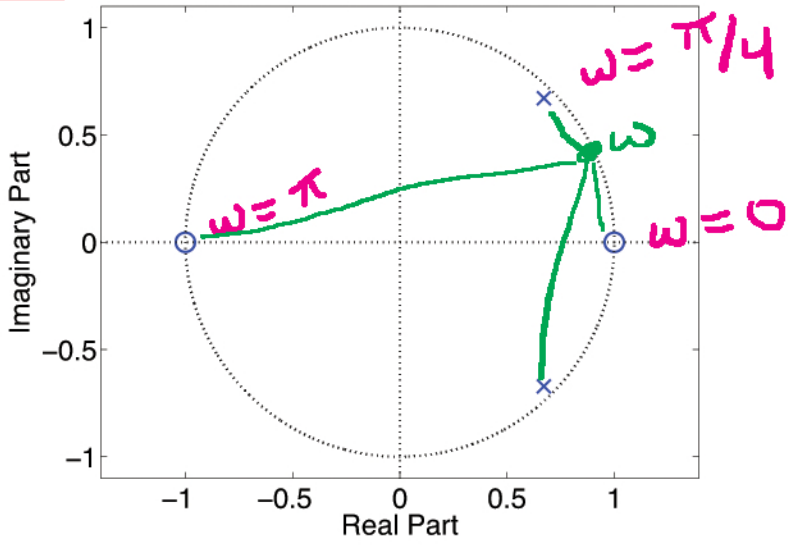
zero at angle ω_0 on circle $\Leftrightarrow$ $|H(e^{j\omega_0})| = 0$

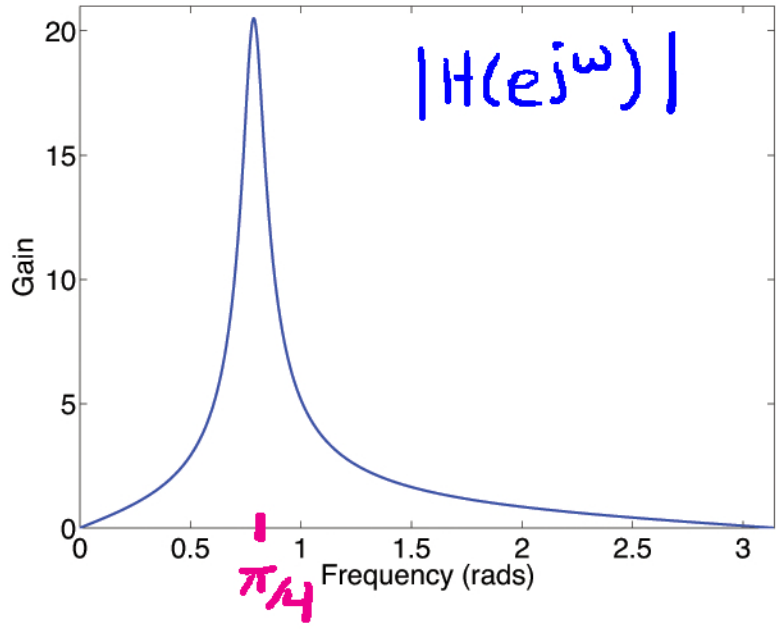


pole at angle ω_0 near circle $\Leftrightarrow$ peak in $|H(e^{j\omega_0})|$



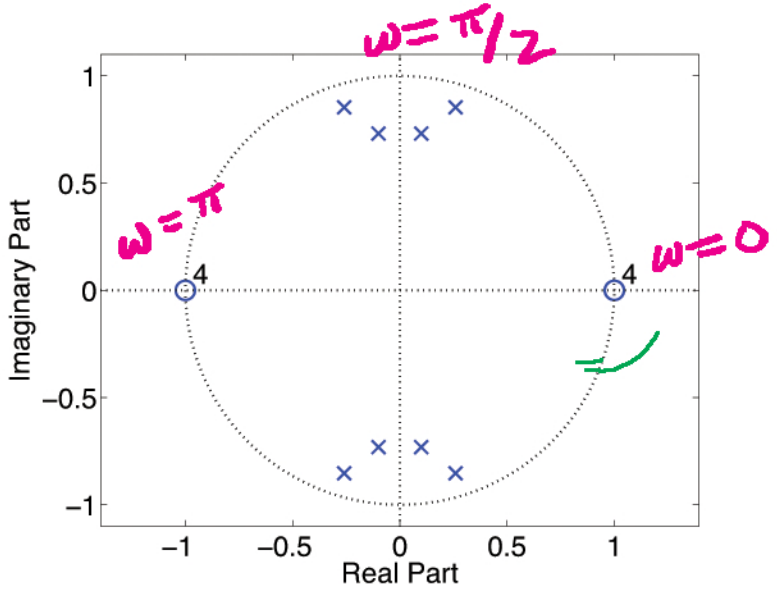
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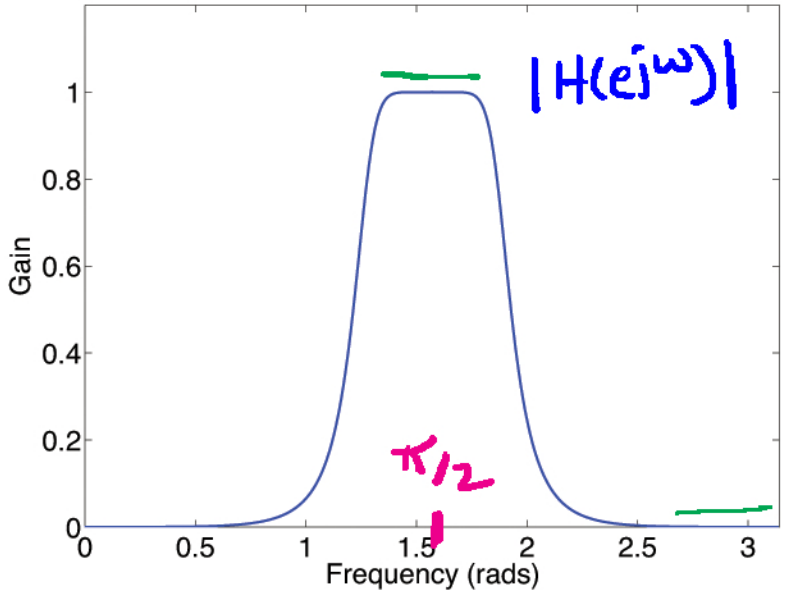




- poles push response up

- zeros pull response down





- Bandpass, lowpass, highpass, etc can be identified from pole/zero locations

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